Category Theory for Mathematical Physics — Exercise Sheet 3

Monoidal categories and string diagrams

Variant: Solutions

Conventions as in Lecture 3: pictures read upward, the first factor on the left; Vec_G^ω and its cocycle condition as in Definition 3.7 and Theorem 3.8. The group $\mathbb{Z}_N$ is written additively, with elements $0, 1, \dots, N-1$ and $[x]_N$ the remainder of an integer x mod N; $\zeta_N = e^{2\pi i/N}$. In Exercise 4, as in Lecture 3, $\mathbb{Z}_2 = \{1, s\}$ is written multiplicatively, and the *semion category* is Vec_G^ω on it with $\omega(s,s,s) = -1$. Here ε is the carry of Exercise 3 (not the toric-code anyon); e and u are the units of two operations in Exercise 2, and u is a unit of $\mathbb{Z}_N$ in Exercise 6; in Exercise 5, φ is a group automorphism (not the golden ratio).

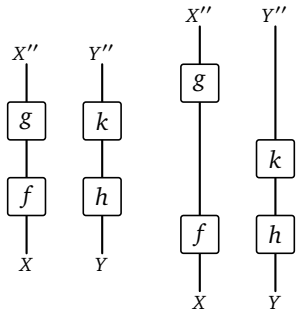
Exercise 1 (Interchange and pictures). Let $f\colon X \to X'$, $g\colon X' \to X''$, $h\colon Y \to Y'$ and $k\colon Y' \to Y''$.

(i) Draw $(g \otimes k) \circ (f \otimes h)$ and $(g \otimes \mathrm{id}_{Y''}) \circ (\mathrm{id}_{X'} \otimes k) \circ (f \otimes h)$, and show that both equal $(g \circ f) \otimes (k \circ h)$.

(ii) In $\mathrm{Mat}_{\mathbb{C}}$ let $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. Compute $A \otimes B$ and $B \otimes A$ (Kronecker products), and check that $(A \otimes I)(I \otimes B) = A \otimes B = (I \otimes B)(A \otimes I)$.

(iii) Boxes on different strands slide past each other (Equation (3.3)). Can two boxes on the same strand? Answer with A and B.

Solution. (i) Both are the picture with f below g on the left strand and h below k on the right:



The first is $(g \otimes k) \circ (f \otimes h) = (g \circ f) \otimes (k \circ h)$ by Theorem 3.4. For the second, interchange twice: $(\mathrm{id}_{X'} \otimes k) \circ (f \otimes h) = f \otimes (k \circ h)$, and then $(g \otimes \mathrm{id}_{Y''}) \circ (f \otimes (k \circ h)) = (g \circ f) \otimes (k \circ h)$. In the second picture g sits higher than k; the heights of boxes on different strands carry no meaning.

(ii) With $A \otimes B = (a_{ij}B)$,

$$A \otimes B = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}, \qquad B \otimes A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{pmatrix}.$$

They differ: the order of the strands matters. By the mixed-product rule $(P \otimes Q)(R \otimes S) = PR \otimes QS$ (Example 3.2, the first condition of Definition 3.1 in $\mathrm{Mat}_{\mathbb{C}}$), $(A \otimes I)(I \otimes B) = A \otimes B = (I \otimes B)(A \otimes I)$; multiplying out the 4×4 matrices confirms it.

(iii) No. Two boxes on one strand compose, and composition need not commute: $AB = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ while $BA = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. Only boxes on different strands slide past each other.

Exercise 2 (Eckmann–Hilton, in full). Let M be a set with two binary operations $\cdot$ and $*$, with units u for $\cdot$ and e for $*$, such that

$$(a * b) \cdot (c * d) = (a \cdot c) * (b \cdot d) \quad \text{for all } a, b, c, d \in M.$$

(i) Show that $u = e$.

(ii) Show that $a \cdot b = a * b$ and $a \cdot b = b \cdot a$ for all a, b.

(iii) Apply this to $\mathrm{End}(1)$ in a strict monoidal category, with $\cdot = \circ$ and $* = \otimes$, to recover Theorem 3.6, and draw the computation of (ii) for $s, t \in \mathrm{End}(1)$ as pictures of boxes with no strands. Where does strictness enter?

Solution. (i) u is the unit of $\cdot$ and e that of $*$, so

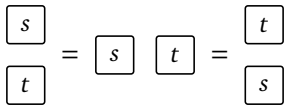
$$u = u \cdot u = (e * u) \cdot (u * e) = (e \cdot u) * (u \cdot e) = e * e = e.$$

(ii) With $u = e$ a unit for both operations,

$$a \cdot b = (a * e) \cdot (e * b) = (a \cdot e) * (e \cdot b) = a * b,$$
$$a \cdot b = (e * a) \cdot (b * e) = (e \cdot b) * (a \cdot e) = b * a = b \cdot a.$$

(The last step uses the first line with a and b swapped.) So the two operations coincide and are commutative.

(iii) In a strict monoidal category, $s \otimes t$ for $s, t \in \mathrm{End}(1)$ is a morphism $1 \otimes 1 \to 1 \otimes 1$, that is $1 \to 1$; so $\otimes$ is an operation on $\mathrm{End}(1)$. The first condition of Definition 3.1 is the interchange law $(s \otimes t) \circ (s' \otimes t') = (s \circ s') \otimes (t \circ t')$, the displayed hypothesis with $\cdot = \circ$ and $* = \otimes$. The unit of $\circ$ is id_1. Strictness enters twice: $1 \otimes 1 = 1$ makes $\otimes$ an operation on $\mathrm{End}(1)$, and in the unit of $\otimes$, r_1 and l_1 are identities, so naturality of r gives $s \otimes \mathrm{id}_1 = s$, and of l gives $\mathrm{id}_1 \otimes s = s$; thus id_1 is the unit of $\otimes$ too. By (ii), $s \circ t = s \otimes t = t \circ s$: Theorem 3.6. In pictures, t below s, then s beside t, then s below t:



which is Equation (3.5). The middle picture is $s \otimes t$; sliding s up past t (Equation (3.3)) gives the left picture, $s \circ t$, and sliding it down gives the right one, $t \circ s$. A general monoidal category is reduced to this case by Theorem 3.3(a), as in the lecture.

Exercise 3 (The cocycles ω_p of $\mathbb{Z}_N$). For $b, c \in \{0, \dots, N-1\}$ let $\varepsilon(b, c) = (b + c - [b+c]_N)/N$, the *carry* of $b + c$, and let $\omega_p(a, b, c) = e^{2\pi i\, p\, a (b+c-[b+c]_N)/N^2}$.

(i) Show that $\varepsilon(b, c) \in \{0, 1\}$ and $\omega_p(a, b, c) = \zeta_N^{\,p\, a\, \varepsilon(b,c)}$. Hence ω_p depends only on p mod N.

(ii) Show that $\varepsilon(b, c) + \varepsilon(a, [b + c]_N) = \varepsilon(a, b) + \varepsilon([a + b]_N, c)$.

(iii) Draw the pentagon of Definition 3.1 for simple objects g, h, k, m of Vec_G^{ω}, with each vertex labelled by its bracketing (each is the object δ_{ghkm}) and each arrow by its scalar, and the triangle for $g, 1, h$ likewise; read off the conditions of Theorem 3.8. Then show that ω_p is normalised and satisfies the cocycle condition, which for $\mathbb{Z}_N$ written additively reads

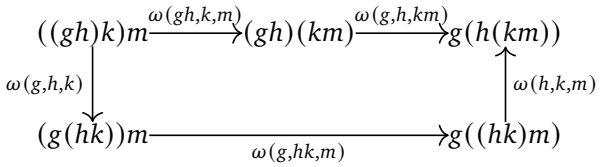
$$\omega(a + b, c, d)\, \omega(a, b, c + d)$$
$$= \omega(a, b, c)\, \omega(a, b + c, d)\, \omega(b, c, d).$$

(iv) For $N = 3$ and $p = 1$, list the triples (a, b, c) with $\omega_1(a, b, c) \neq 1$, and the values there.

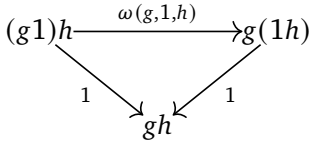
Solution. (i) Since $0 \leq b+c \leq 2N-2$, $[b+c]_N$ is $b+c$ or $b+c-N$, so $\varepsilon(b, c) \in \{0, 1\}$, with $\varepsilon = 1$ exactly when $b+c \geq N$. Then $p\,a(b+c-[b+c]_N)/N^2 = p\,a\,\varepsilon(b, c)/N$, so $\omega_p(a, b, c) = \zeta_N^{p\,a\,\varepsilon(b,c)}$. Since $\zeta_N^N = 1$, replacing p by $p+N$ changes nothing.

(ii) $b + c = [b + c]_N + N\varepsilon(b, c)$ and $a + [b + c]_N = [a + b + c]_N + N\varepsilon(a, [b + c]_N)$, so $a + b + c = [a + b + c]_N + N\big(\varepsilon(b, c) + \varepsilon(a, [b + c]_N)\big)$. Adding in the other order gives $a + b + c = [a + b + c]_N + N\big(\varepsilon(a, b) + \varepsilon([a + b]_N, c)\big)$. Compare.

(iii) In the pentagon every object is δ_{ghkm} and every arrow is a scalar times the identity ($\otimes\,\mathrm{id}$ does not change the scalar):



so it commutes exactly when $\omega(gh, k, m)\omega(g, h, km) = \omega(g, h, k)\omega(g, hk, m)\omega(h, k, m)$. In the triangle every vertex is δ_{gh}, the arrow $\alpha_{g,1,h}$ carries $\omega(g, 1, h)$ and the two unitor sides carry 1:



so it commutes exactly when $\omega(g, 1, h) = 1$. Normalised: $\varepsilon(0, c) = 0$, so $\omega_p(a, 0, c) = 1$ (and $\omega_p = 1$ also when $a = 0$ or $c = 0$). For the cocycle condition, the exponents of ζ_N^p are, with all sums reduced mod N inside the arguments,

$$\text{left: } [a + b]_N\,\varepsilon(c, d) + a\,\varepsilon(b, [c + d]_N),$$
$$\text{right: } a\,\varepsilon(b, c) + a\,\varepsilon([b + c]_N, d) + b\,\varepsilon(c, d).$$

By (ii) for (b, c, d), $\varepsilon(b, c) + \varepsilon([b + c]_N, d) = \varepsilon(c, d) + \varepsilon(b, [c + d]_N)$, so the right side is $(a + b)\varepsilon(c, d) + a\,\varepsilon(b, [c + d]_N)$. The difference, right minus left, is $(a + b - [a + b]_N)\varepsilon(c, d) = N\varepsilon(a, b)\varepsilon(c, d)$, and $\zeta_N^{pN\varepsilon(a,b)\varepsilon(c,d)} = 1$. So the two sides agree.

(iv) $\omega_1(a, b, c) = \zeta_3^{a\,\varepsilon(b,c)} \neq 1$ exactly when $a \neq 0$ and $b + c \geq 3$, that is $a \in \{1, 2\}$ and $(b, c) \in \{(1, 2), (2, 1), (2, 2)\}$: six triples, with value ζ_3^a.

Exercise 4 (The forgetful functor as a monoidal functor). (i) Show that the forgetful functor $U\colon \mathrm{Rep}(G) \to \mathrm{Vect}_{\mathrm{fd}}$, with J and J_0 identities, is monoidal (Definition 3.10).

(ii) Let $F\colon \mathrm{Vec}_G^\omega \to \mathrm{Vect}_{\mathrm{fd}}$ send a graded space to its underlying space, so $F(\delta_g) = \mathbb{C}$, and give it $J_{g,h} = j(g, h)\,\mathrm{id}_{\mathbb{C}}$ (with $\mathbb{C} \otimes \mathbb{C} = \mathbb{C}$) and $J_0 = j_0\,\mathrm{id}_{\mathbb{C}}$, for nonzero numbers $j(g, h)$, j_0. Show that (F, J, J_0) is monoidal exactly when $\omega = \delta j$, with δj as in Lecture 3, and $j(1, g) = j(g, 1) = j_0^{-1}$.

(iii) Show that for $G = \mathbb{Z}_2$ and $\omega(s, s, s) = -1$ no such (F, J, J_0) exists.

Solution. (i) In $\mathrm{Rep}(G)$ the associator and unitors are those of Vect (Example 3.2), and U does not change morphisms. With J and J_0 identities, the first condition of Definition 3.10 reads $U(\alpha_{X,Y,Z}) = \alpha_{UX,UY,UZ}$, and the unit conditions read $U(l_X) = l_{UX}$, $U(r_X) = r_{UX}$: all true. J is natural because intertwiners are linear maps and $\otimes$ is that of Vect.

(ii) F is linear and sends δ_g to $\mathbb{C}$, so on the simple objects every morphism in the conditions is a scalar times $\mathrm{id}_{\mathbb{C}}$, and it suffices to check them there (both sides are natural, hence

determined on simple objects, as in the proof of Theorem 3.8). $F(\alpha_{g,h,k}) = \omega(g,h,k)$, Vect's associator on $\mathbb{C}\otimes\mathbb{C}\otimes\mathbb{C}$ is 1, and $J\otimes\mathrm{id}$ has the scalar of J. The first condition becomes

$$\omega(g,h,k)\,j(gh,k)\,j(g,h) = j(h,k)\,j(g,hk),$$

that is $\omega(g,h,k) = j(h,k)\,j(g,hk)/\big(j(gh,k)\,j(g,h)\big) = (\delta j)(g,h,k)$. The unitors of Vec_G^ω and of Vect on $\mathbb{C}$ are identities, so the unit conditions read $j(1,g)\,j_0 = 1$ and $j(g,1)\,j_0 = 1$. Conversely, if both hold, every condition holds on simple objects and hence everywhere.

(iii) At (s,s,s), with $s\cdot s = 1$, $(\delta j)(s,s,s) = j(s,s)\,j(s,1)/\big(j(1,s)\,j(s,s)\big) = j(s,1)/j(1,s)$, which is 1 because $j(1,s) = j(s,1) = j_0^{-1}$. So $\omega = \delta j$ would force $\omega(s,s,s) = 1$. For $\omega(s,s,s) = -1$ no (F,J,J_0) exists: the semion category has no monoidal functor to $\mathrm{Vect}_{\mathrm{fd}}$ that forgets the grading. (This is the computation of Proposition 3.11 with $\omega' = 1$.)

Exercise 5 (Monoidal equivalences of Vec_G^ω). Let ω and ω' be normalised 3-cocycles on a finite group G. You may use that an equivalence is full, faithful and essentially surjective (Leinster Prop. 1.3.18, the converse of Proposition 1.17).

(i) Let φ be an automorphism of G and $j\colon G\times G\to\mathbb{C}^\times$ satisfy

$$\begin{aligned}&\omega'(\varphi(g),\varphi(h),\varphi(k))\\ &\quad= \omega(g,h,k)\,\frac{j(gh,k)\,j(g,h)}{j(h,k)\,j(g,hk)}\end{aligned}\tag{$*$}$$

for all g,h,k. Show that $F(\delta_g) = \delta_{\varphi(g)}$, extended to direct sums, with $J_{g,h} = j(g,h)$ and $J_0 = j(1,1)^{-1}$, is a monoidal equivalence $\mathrm{Vec}_G^\omega\to\mathrm{Vec}_G^{\omega'}$.

(ii) Conversely, let (F,J,J_0) be a monoidal equivalence $\mathrm{Vec}_G^\omega\to\mathrm{Vec}_G^{\omega'}$. Show that there are an automorphism φ and a function j satisfying $(*)$.

(iii) Conclude the classification stated in Lecture 3: Vec_G^ω and $\mathrm{Vec}_G^{\omega'}$ are monoidally equivalent exactly when ω' is obtained from ω by an automorphism of G and a coboundary.

Solution. (i) F sends δ_g to $\delta_{\varphi(g)}$ and a morphism between simple objects, a scalar (or 0 between different ones), to the same scalar; extended to direct sums by linearity it is a linear functor. It is full and faithful, because φ is a bijection and the Hom spaces between simple objects are $\mathbb{C}$ or 0 on both sides in the same places, and essentially surjective, since $\delta_k = F(\delta_{\varphi^{-1}(k)})$; by Proposition 1.17 it is an equivalence. As $\varphi(g)\varphi(h) = \varphi(gh)$, $J_{g,h} = j(g,h)$ is an isomorphism $F(\delta_g)\otimes F(\delta_h) = \delta_{\varphi(gh)} = F(\delta_{gh})$, extended to direct sums; it is natural because morphisms are scalars on simple summands. On simple objects g,h,k the first condition of Definition 3.10 is the scalar equation

$$\omega(g,h,k)\,j(gh,k)\,j(g,h) = j(h,k)\,j(g,hk)\,\omega'(\varphi(g),\varphi(h),\varphi(k)),$$

which is $(*)$. The unitors are identities, so the unit conditions read $j(1,g)\,j(1,1)^{-1} = 1 = j(g,1)\,j(1,1)^{-1}$. With ω,ω' normalised and $\varphi(1) = 1$, $(*)$ at $(g,1,h)$ reads $1 = j(g,h)\,j(g,1)/\big(j(1,h)\,j(g,h)\big)$, so $j(g,1) = j(1,h)$ for all g,h: both equal $j(1,1)$, and the unit conditions hold.

(ii) F is full and faithful, so $\mathrm{End}(F(X))\cong\mathrm{End}(X)$; by Theorem 2.13 an object of a semisimple category has $\mathrm{End}\cong\mathbb{C}$ exactly when it is isomorphic to a simple object. So each $F(\delta_g)$ is isomorphic to exactly one $\delta_{\varphi(g)}$ (the δ_k are pairwise non-isomorphic). The map φ is injective, since a full and faithful functor reflects isomorphisms (an isomorphism $F(A)\to F(B)$ is $F(a)$, its inverse $F(b)$, and faithfulness gives $ba = \mathrm{id}$, $ab = \mathrm{id}$); and surjective, since $\delta_k\cong F(X)$ for some X, which then has $\mathrm{End}(X)\cong\mathbb{C}$ and is some δ_g. Choose isomorphisms $u_g\colon F(\delta_g)\to\delta_{\varphi(g)}$. Then $u_{gh}\circ J_{g,h}\circ(u_g^{-1}\otimes u_h^{-1})$ is an isomorphism $\delta_{\varphi(g)\varphi(h)}\to\delta_{\varphi(gh)}$,

so $\varphi(g)\varphi(h) = \varphi(gh)$: φ is an automorphism, and this isomorphism is a scalar $j(g,h)$. Conjugating the first condition of Definition 3.10 by the u's turns every arrow into a scalar: $F(\alpha_{g,h,k}) = \omega(g,h,k)$ id since F is linear, the J's become j's, and α' is $\omega'(\varphi(g),\varphi(h),\varphi(k))$. The resulting scalar equation is $(*)$.

(iii) Rearranged, $(*)$ says $\omega' \circ (\varphi\times\varphi\times\varphi) = \omega\,(\delta j)^{-1}$, with δj as in Lecture 3: up to the coboundary of j, ω' is ω moved by the automorphism φ^{-1} in the sense of Exercise 6, $\omega'(x,y,z) = \big(\omega\,(\delta j)^{-1}\big)(\varphi^{-1}x,\varphi^{-1}y,\varphi^{-1}z)$. By (i) and (ii) this holds exactly when the two categories are monoidally equivalent: the classes of Vec_G^ω are the classes $[\omega]\in H^3(G,\mathbb{C}^\times)$ up to automorphisms of G, the statement Lecture 3 cites from EGNO Prop. 2.6.1. With $\varphi = \mathrm{id}$ this is Proposition 3.11's computation.

Exercise 6 ⋆. *The classes of* $\mathrm{Vec}_{\mathbb{Z}_N}^\omega$ *up to equivalence.* For a normalised 3-cocycle ω on $\mathbb{Z}_N$ and $a\in\mathbb{Z}_N$ let

$$Q_\omega(a) = \prod_{r=0}^{N-1} \omega(a,ra,a),$$

with ra reduced mod N. Two cocycles are *cohomologous* if $\omega' = \omega\,(\delta j)^{-1}$ for some $j\colon \mathbb{Z}_N^2\to\mathbb{C}^\times$ (Lecture 3).

(i) Show that $Q_{\omega'} = Q_\omega$ when ω' and ω are cohomologous.

(ii) Show that $Q_{\omega_p}(1) = \zeta_N^p$. Conclude that $\omega_0,\dots,\omega_{N-1}$ are pairwise not cohomologous.

(iii) For a unit u mod N let $(u^*\omega)(a,b,c) = \omega(ua,ub,uc)$, the cocycle moved by the automorphism $a\mapsto ua$. Show that $Q_{u^*\omega}(1) = Q_\omega(u)$ and $Q_{\omega_p}(u) = \zeta_N^{pu^2}$, with u taken in $\{1,\dots,N-1\}$.

(iv) You may use that every normalised 3-cocycle on $\mathbb{Z}_N$ is cohomologous to some ω_p (EGNO Ex. 2.6.4). With Exercise 5, show that $\mathrm{Vec}_{\mathbb{Z}_N}^{\omega_p}$ and $\mathrm{Vec}_{\mathbb{Z}_N}^{\omega_{p'}}$ are monoidally equivalent exactly when $p'\equiv pu^2$ mod N for some unit u, and count the classes for $N = 2,3,4,5$.

Solution. (i) Write the group additively, so $\delta j(a,b,c) = j(b,c)\,j(a,b+c)/\big(j(a+b,c)\,j(a,b)\big)$. At (a,ra,a):

$$(\delta j)(a,ra,a) = \frac{j(ra,a)}{j((r+1)a,a)}\cdot\frac{j(a,(r+1)a)}{j(a,ra)}.$$

As r runs over $0,\dots,N-1$, $(r+1)a$ runs over the same multiples of a as ra (since $Na=0$), so the product of each fraction over r is 1. Hence $Q_{\omega(\delta j)^{-1}}(a) = Q_\omega(a)\big/\prod_r(\delta j)(a,ra,a) = Q_\omega(a)$.

(ii) By Exercise 3(i), $\omega_p(1,r,1) = \zeta_N^{p\,\varepsilon(r,1)}$, and $\varepsilon(r,1) = 1$ only for $r = N-1$. So $Q_{\omega_p}(1) = \zeta_N^p$. If ω_p and $\omega_{p'}$ were cohomologous, (i) would give $\zeta_N^p = \zeta_N^{p'}$, so $p = p'$ for $p,p'\in\{0,\dots,N-1\}$.

(iii) $Q_{u^*\omega}(1) = \prod_r \omega(u,ru,u) = Q_\omega(u)$. For ω_p, $\omega_p(u,[ru]_N,u) = \zeta_N^{p\,u\,\varepsilon([ru]_N,u)}$, and $\varepsilon([ru]_N,u) = 1$ exactly when $[ru]_N\geq N-u$. Since u is a unit, $[ru]_N$ runs over all of $\{0,\dots,N-1\}$ once, and u of these values are $\geq N-u$. So $Q_{\omega_p}(u) = \zeta_N^{pu\cdot u} = \zeta_N^{pu^2}$.

(iv) By Exercise 5, the two categories are equivalent exactly when $\omega_{p'}$ is cohomologous to $u^*\omega_p$ for an automorphism $a\mapsto ua$ (every automorphism of $\mathbb{Z}_N$ is of this form, u a unit; Exercise 5's $(*)$ moves ω by φ^{-1}, and u and u^{-1} run over the same units). By the cited fact $u^*\omega_p$ is cohomologous to some ω_q, and by (i) and (iii) $\zeta_N^q = Q_{u^*\omega_p}(1) = Q_{\omega_p}(u) = \zeta_N^{pu^2}$, so $q\equiv pu^2$; by (ii) the class of ω_q fixes q. So the classes are the orbits of $p\mapsto pu^2$. For $N=2$ and $N=3$ every unit square is 1 ($2^2 = 4\equiv 1$ mod 3): 2 and 3 classes. For $N=4$ the units $1,3$ have squares $1,9\equiv 1$: 4 classes. For $N=5$ the unit squares are 1 and $4\equiv -1$, so $p\sim -p$: the classes are $\{0\}$, $\{1,4\}$ and $\{2,3\}$, three in all.